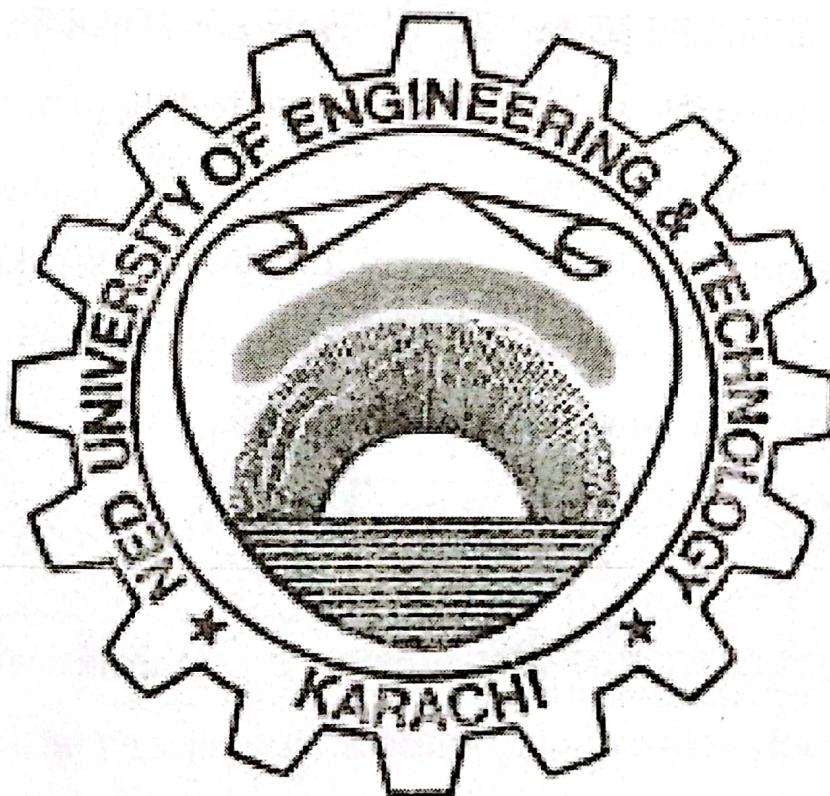


Practical Workbook
CS-251
Logic Design and Switching Theory



PROCEDURE

1. Set the power supply to 5V. With the help of a multimeter check the voltage at the output knobs of the power supply.
2. Connect wires, long enough to reach the bread board, with the two knobs of the power supply. Again using multimeter, check the voltage at the non-connected end of the wires.
3. Insert the 7408 quad 2 input AND gate IC on to the bread board and make supply and ground connections by joining 5V wire to pin # 14 and 0V wire to pin # 7.
4. Consult IC's internal connection diagram for input and output pins of the first AND gate. Connect input pins to logic 0 (0V) and observe the output using LED or logic probe.
5. Try different combinations of logic levels at the two inputs. Again observe the output.
6. Repeat the last two steps for all other gates of the same IC. Record the observations.
7. Repeat this procedure for all other ICs.

OBSERVATIONS

AND Gate

A	B	Expected Output	Observed Output
0	0	0	0
0	1	0	0
1	0	0	0
1	1	1	1

OR Gate

A	B	Expected Output	Observed Output
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	1

NOT Gate

A	Expected Output	Observed Output
0	1	1
1	0	0

NAND Gate

A	B	Expected Output	Observed Output
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

NOR Gate

A	B	Expected Output	Observed Output
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

FINDING EXPRESSION FOR GIVEN LOGIC DIAGRAM

LOGIC DIAGRAM

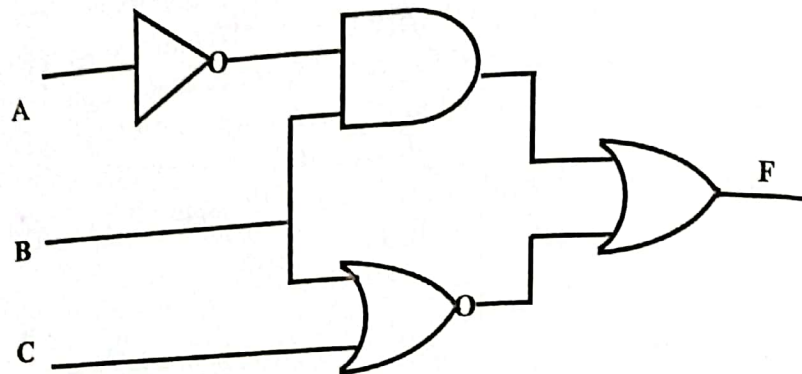


Figure 2: Logic Diagram

PROCEDURE

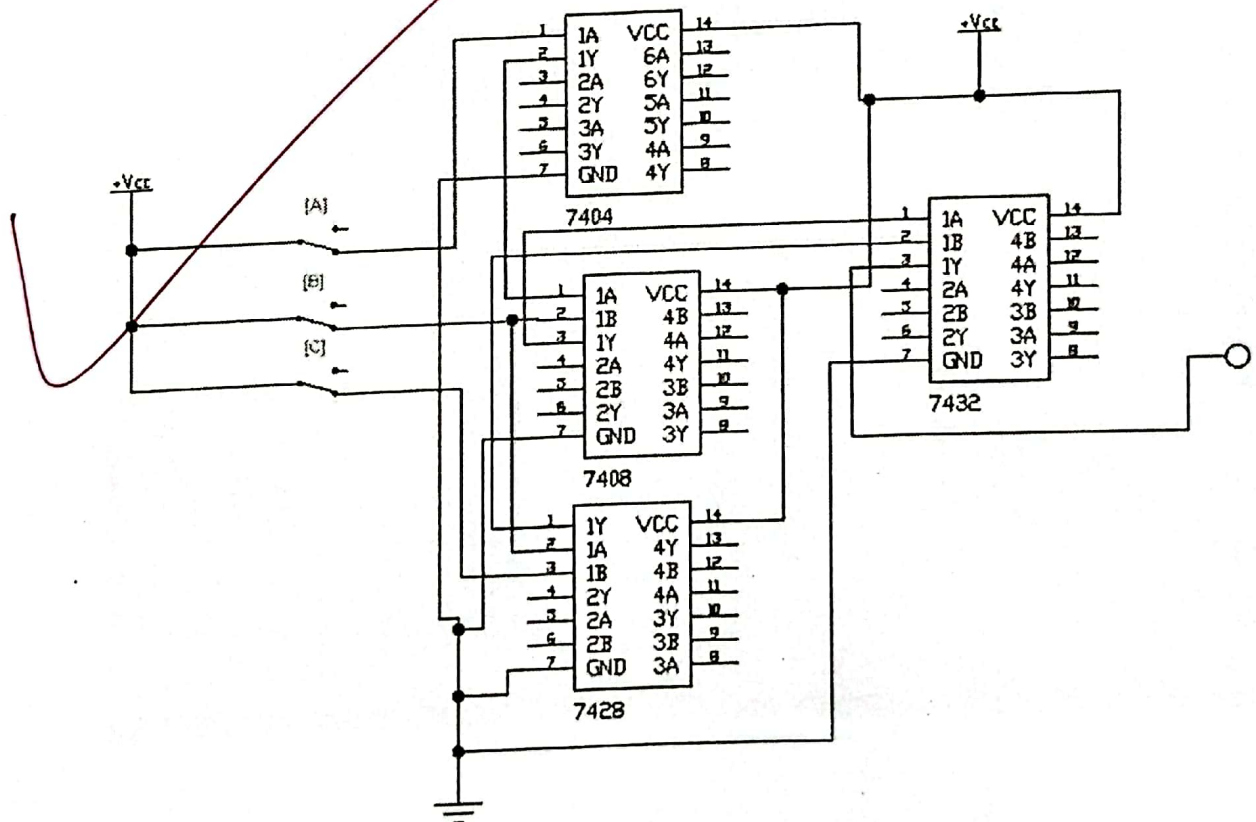
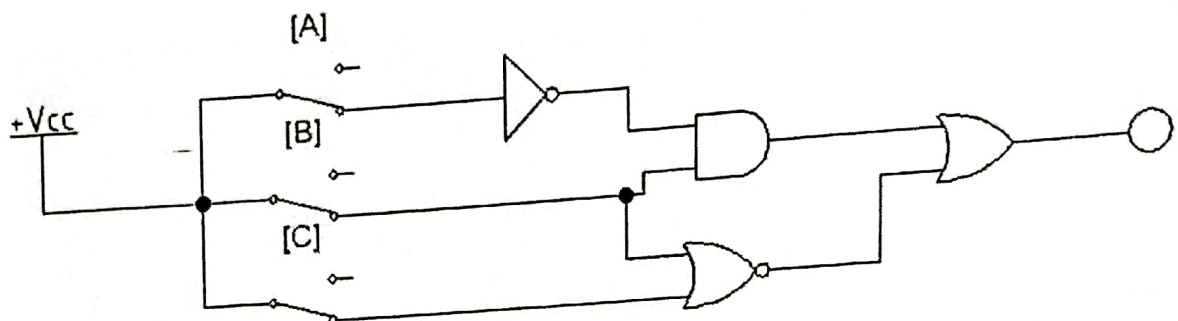
1. Set the power supply to 5V. With the help of a multimeter check the voltage at the output knobs of the power supply.
2. Connect wires, long enough to reach the bread board, with the two knobs of the power supply. Again using millimeter, check the voltage at the non-connected end of the wires.
3. Insert ICs on the bread board and make their supply and ground connections.
4. As given in the logic diagram, make connections using wires and gates in the ICs.
5. Apply different combinations at the three inputs and observe the output.

OBSERVATIONS

Logic expression for the given logic diagram: $(\bar{A} \cdot B) + (B + C)$

A	B	C	Expected Output	Observed Output
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

Practical No.1(gates and IC):



6. Drag and drop V_{cc} and Ground from the Sources Parts Bin.
7. Connect V_{cc} terminal to one end and Ground terminal to the other end of all the switches.
8. Label the circuit properly using text boxes found in the miscellaneous parts bin.
9. Run the circuit using the Activate Simulation switch. Use the keys you have assigned to the switches to toggle them between V_{cc} and Ground connections, thus providing 1 or 0 respectively to the inputs. Record the results as indicated by the probe for all possible combinations of 1s and 0s at the inputs.

EWB Circuit

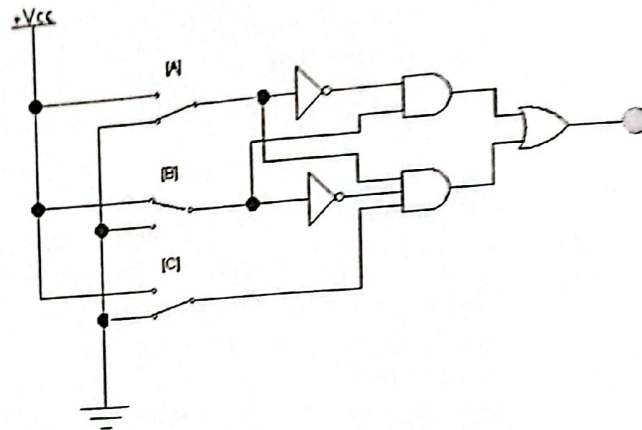


Figure 1: Designed circuit on EWB

Observations

A	B	C	Expected Output	Observed Output
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	1	1
1	1	0	0	0
1	1	1	0	0

USING LOGIC CONVERTER

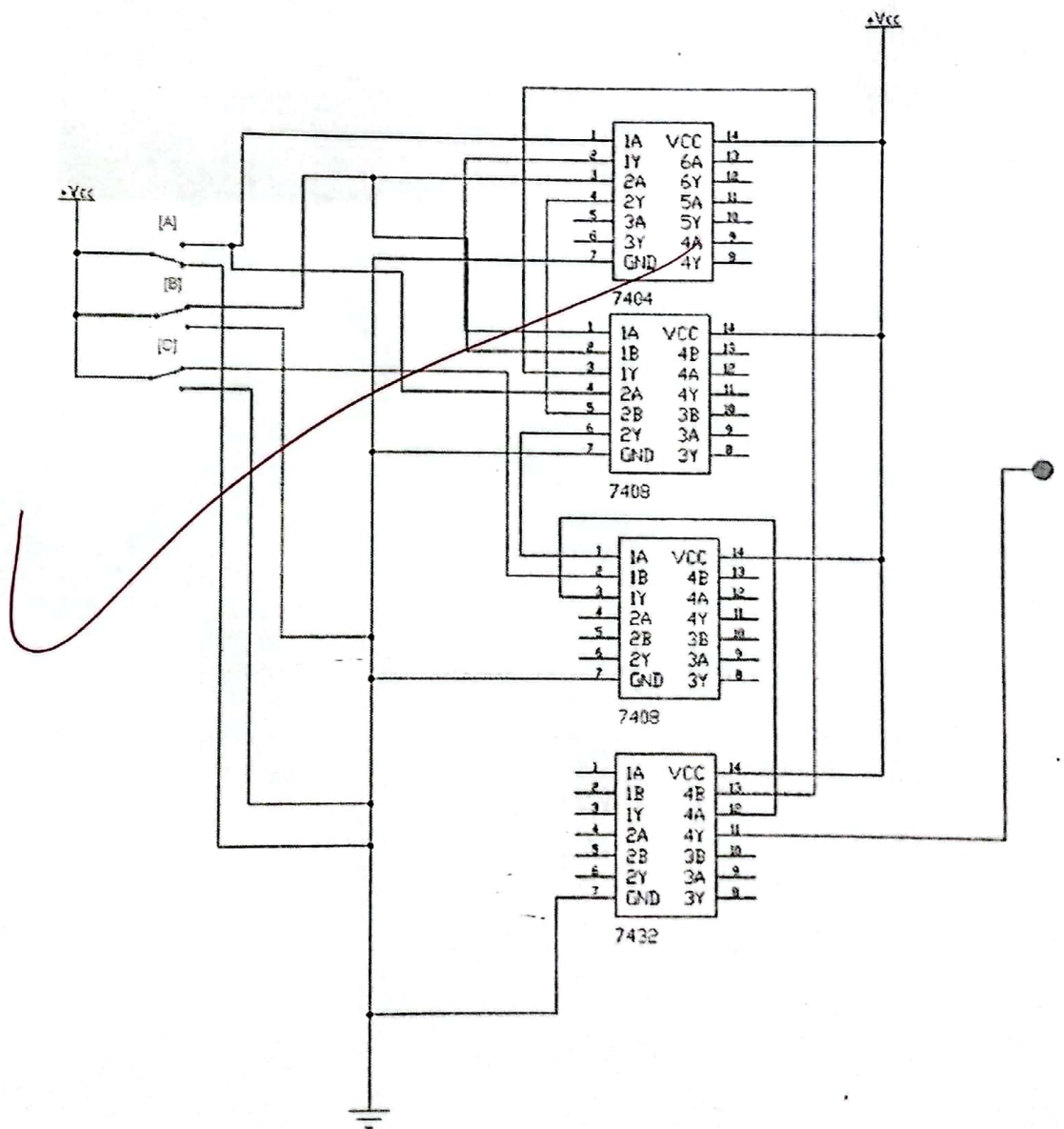
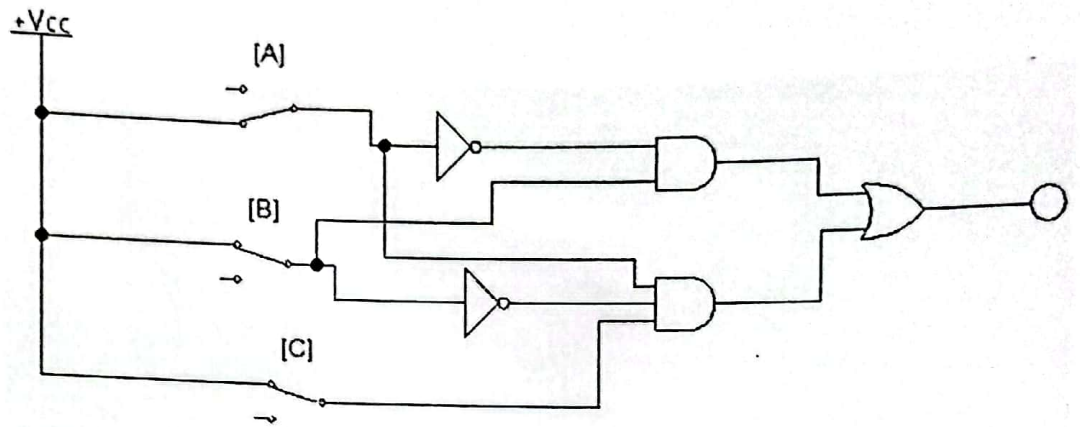
Logic converter can be used to derive truth table and logic expression for a given logic circuit or vice versa, i.e. if a logic circuit is expressed in any one of the three ways, other two can be directly obtained using the logic converter.

From the Instruments Parts Bin, drag and drop Logic Converter on the design area. Double click the Logic Converter to reveal Logic Converter dialog box. This dialog box shows various conversion options between truth table, logic expression and logic circuit.

Finding truth table and logic expression for a given logic circuit

1. Create any arbitrary logic circuit on the design area.
2. Attach the input terminals of the logic converter to the input points in the circuit.

Practical No.2(gates and IC):



Lab Session 03

OBJECTIVE

Simplifying and implementing the given logic expression in hardware and realizing its NAND equivalent.

COMPONENTS REQUIRED

1. Digital logic trainer board or the following components:
Bread board, 5 V - Power Supply, Multimeter, LEDs with Resistors, Switches
2. Logic probe
3. Connecting wires
4. Following ICs and their datasheets or pin configurations:
5. 7400 quad 2 input NAND gate
6. 7404 hex inverter
7. 7408 quad 2 input AND gate
8. 7432 quad 2 input OR gate

GIVEN LOGIC EXPRESSION

$$F(A, B, C, D) = \sum (0, 2, 3, 7, 8, 10, 11, 14, 15)$$

PROCEDURE

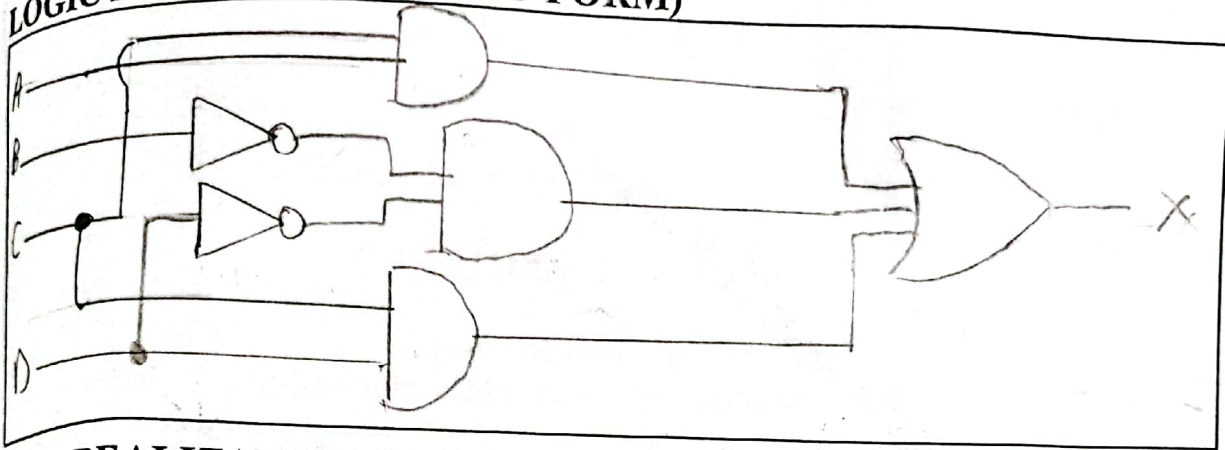
1. Use Karnaugh map to reduce the given function.
2. Draw the circuit diagram for the obtained reduced function.
3. Implement the reduced circuit using digital ICs on a bread board (refer to appendix A for IC pin configurations) and record the observations.
4. Find NAND realization for the simplified circuit.
5. Implement the all NAND circuit using digital ICs on a bread board (refer to appendix A for the pin diagram) and verify the observations taken in step 3.

REDUCTION OF LOGIC EXPRESSION

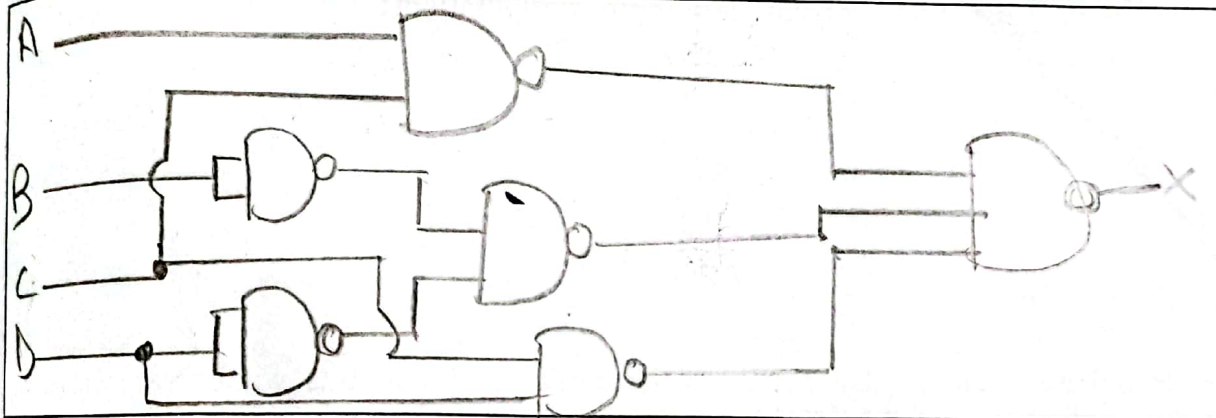
	00	01	11	10
00	1	0	1	1
01	0	0	1	0
11	0	0	1	1
10	1	0	1	1

Reduced logic expression comes out to be: $\bar{B}\bar{D} + CD + AC$

LOGIC DIAGRAM (REDUCED FORM)



NAND REALIZATION OF THE REDUCED LOGIC EXPRESSION



OBSERVATIONS

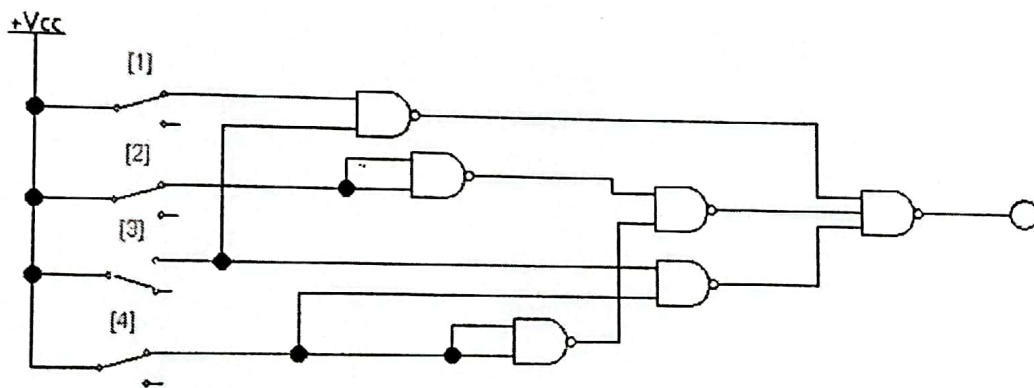
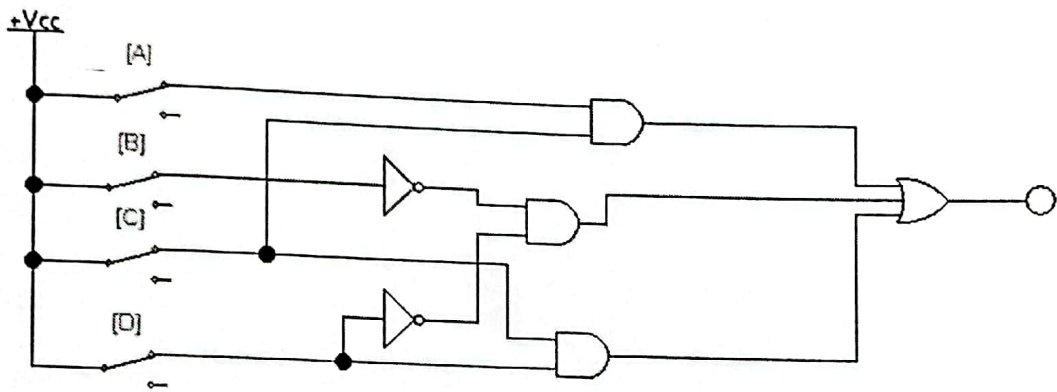
A	B	C	D	Expected Output	Observed Output	
					For simplified expression	For NAND realization
0	0	0	0	1	1	1
0	0	0	1	0	0	0
0	0	1	0	1	1	1
0	0	1	1	1	1	1
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	0
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	0	1	0	0	0
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	0	0	0
1	1	0	1	0	0	0
1	1	1	0	1	1	1
1	1	1	1	1	1	1

EXERCISE

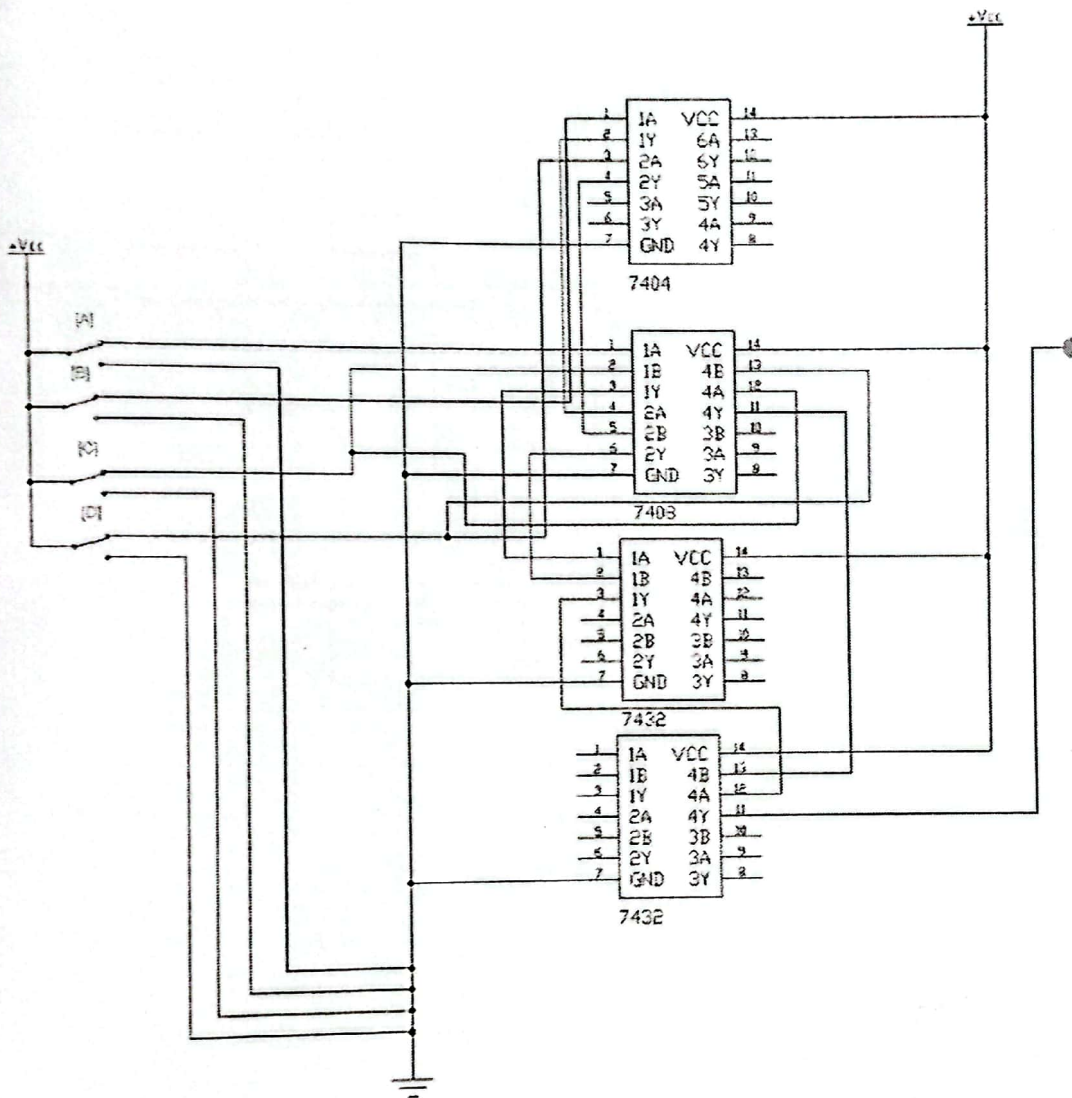
1. Simulate the simplified logic expression and observe any output.

Practical No.3(gates and IC):

1. Simplified expression & NAND Realization Circuits:



2. Simplified expression & NAND Realization ICs:



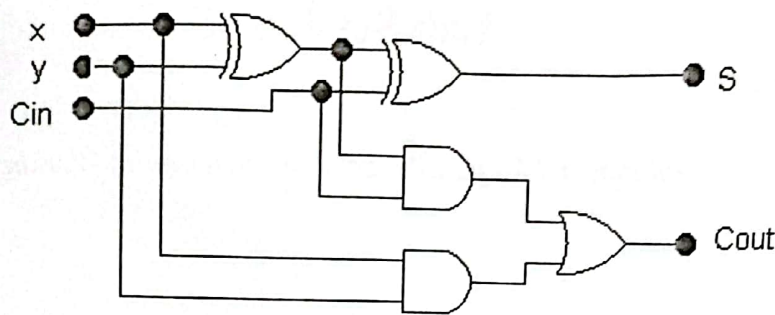


Figure 2: Circuit diagram for Full Adder

HARDWARE IMPLEMENTATION AND OBSERVATIONS

Implement the half adder and full adder circuits in hardware (refer to appendix A for IC pin configurations) and record the observations in the following tables:

Half Adder

Inputs		Outputs	
X	Y	Carry	Sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

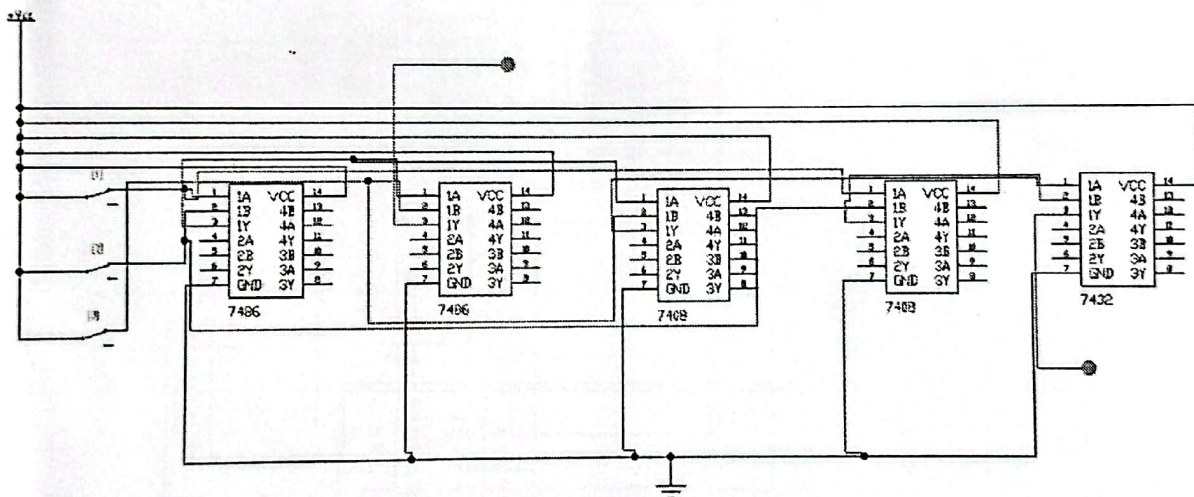
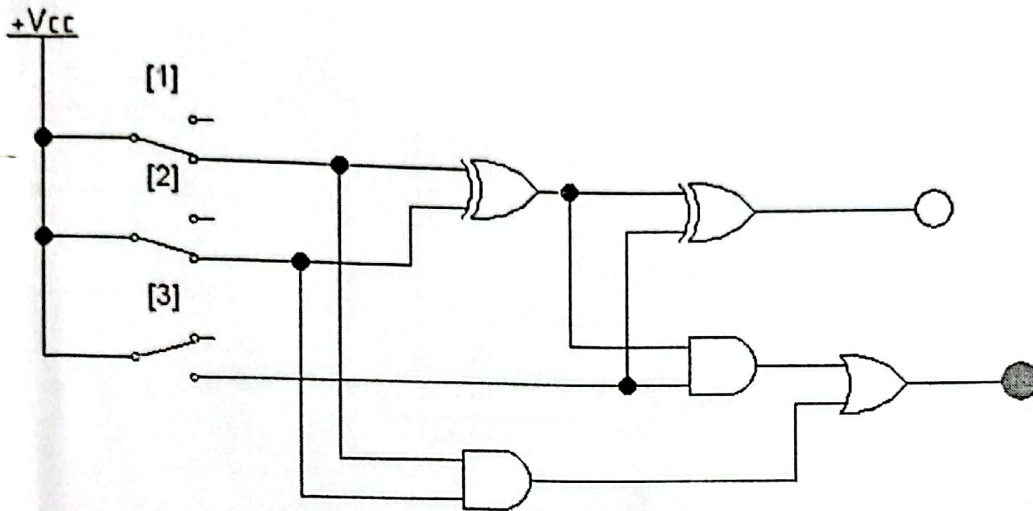
Full Adder

Inputs			Outputs	
X	y	z	Carry	Sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

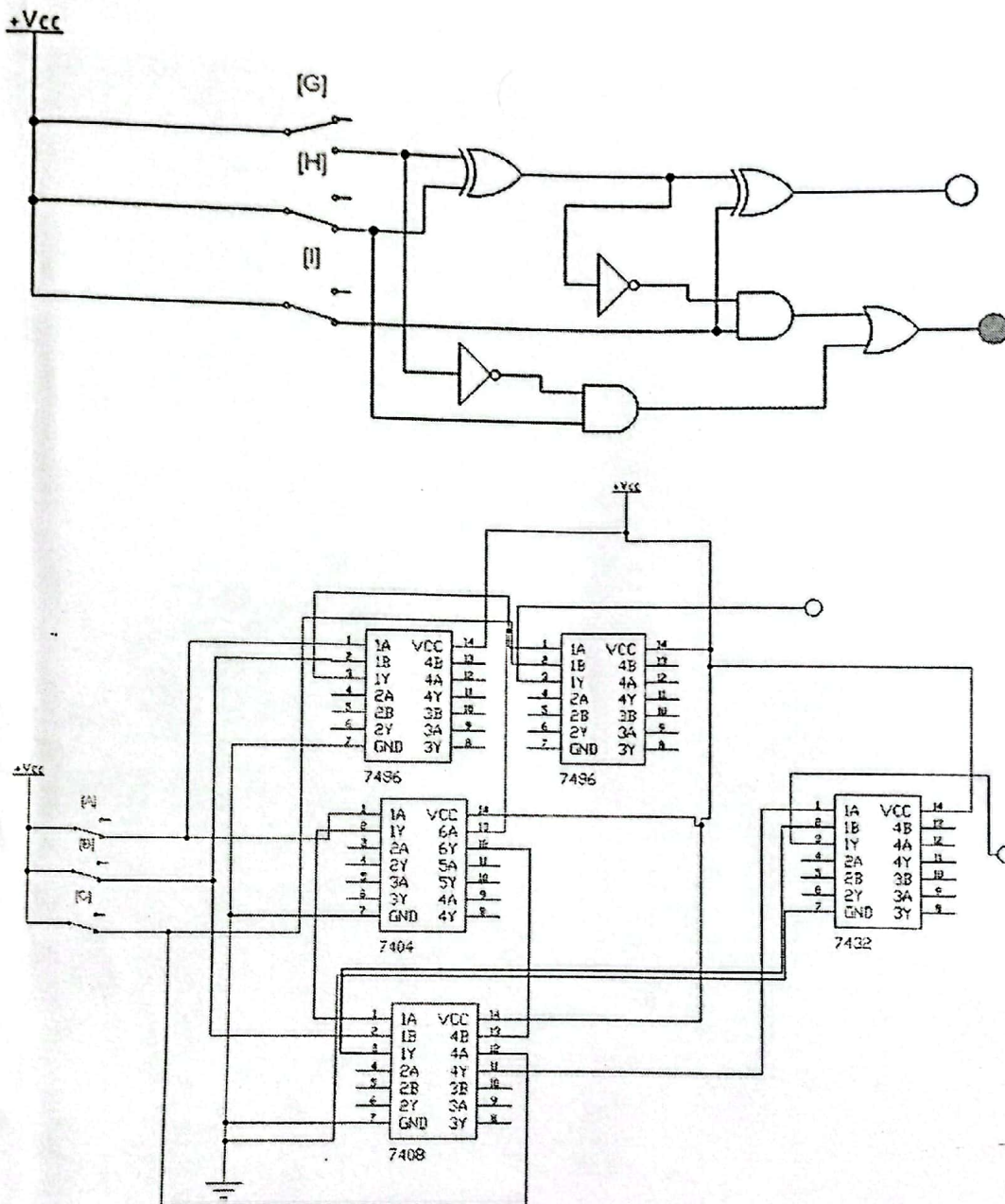
EXERCISES

- Design a full subtractor circuit and simulate it using Electronic Workbench or any simulation tool. Compare the results of simulation with that of your designed circuit. Attach hardcopy of the simulated circuit/output here.

2. Full Adder



3. Full Subtractor



Lab Session 05

OBJECTIVE

Simulating a 4-bit parallel adder on Electronics Workbench

THEORY

4-Bit Parallel Adder

A 4-bit parallel adder can add two 4-bit values. One way of constructing such an adder is to simply cascade four full adders as shown in figure 1. Here $A_3A_2A_1A_0$ and $B_3B_2B_1B_0$ are two 4-bit values to be added. FA_3 , FA_2 , FA_1 and FA_0 are the four full adders that are cascaded in parallel to provide the desired result. $S_3S_2S_1S_0$ is the final 4-bit sum. C_0 is the external carry input (if any) to the circuit and C_4 is the final carry generated by the 4-bit addition.

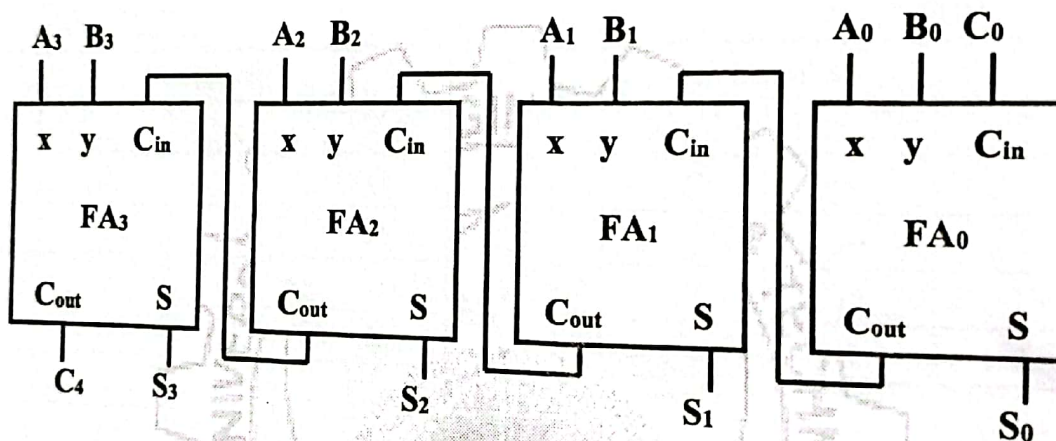


Figure 1: 4-bit Parallel Adder

Creating Subcircuits in Electronic Workbench

Creating a subcircuit allows you to reuse the circuit multiple times in a design and in future designs. Subcircuits may contain basic circuit elements or other subcircuit definitions. To create a subcircuit in Electronics Workbench:

1. First you have to design the circuit that you want to implement. Let's say you want to create a half-adder, so you simply draw a circuit with a XOR and AND gate (XOR SUM and AND is Carry Out).
2. Select all components of the design that needs to be included.
3. From the menu *Circuit*, select *Create Subcircuit*.

EXERCISES

1. Find the sum and output carry for the addition of following two 4-bit numbers if the input carry $C_0 = 0$.

a. $A_3A_2A_1A_0 = 1100$

$B_3B_2B_1B_0 = 1101$

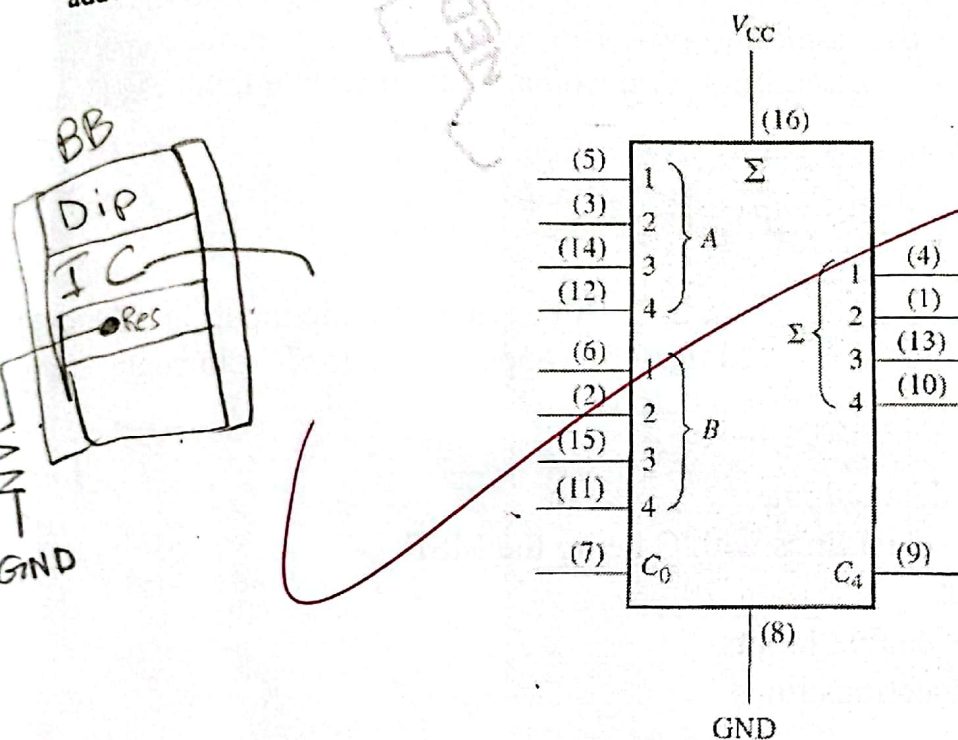
C_4	S_3	S_2	S_1	S_0

b. $A_3A_2A_1A_0 = 1011$

$B_3B_2B_1B_0 = 1010$

C_4	S_3	S_2	S_1	S_0

2. Using the logical symbol of parallel adder (given below) realize 2's complement adder/subtractor unit.



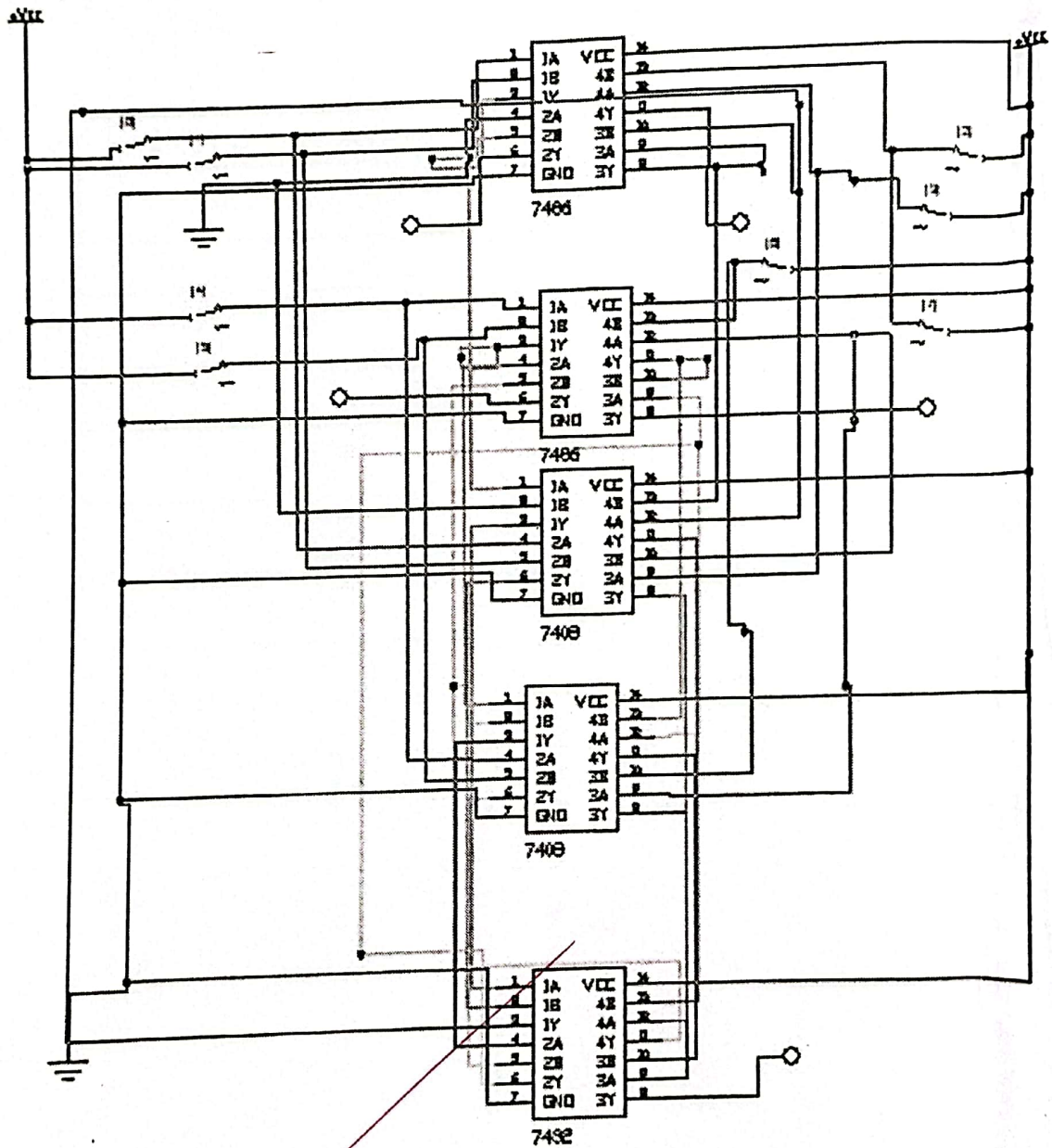
3. Test the designed circuit in Q2 for the given values of A and B when $C_0 = 0$.

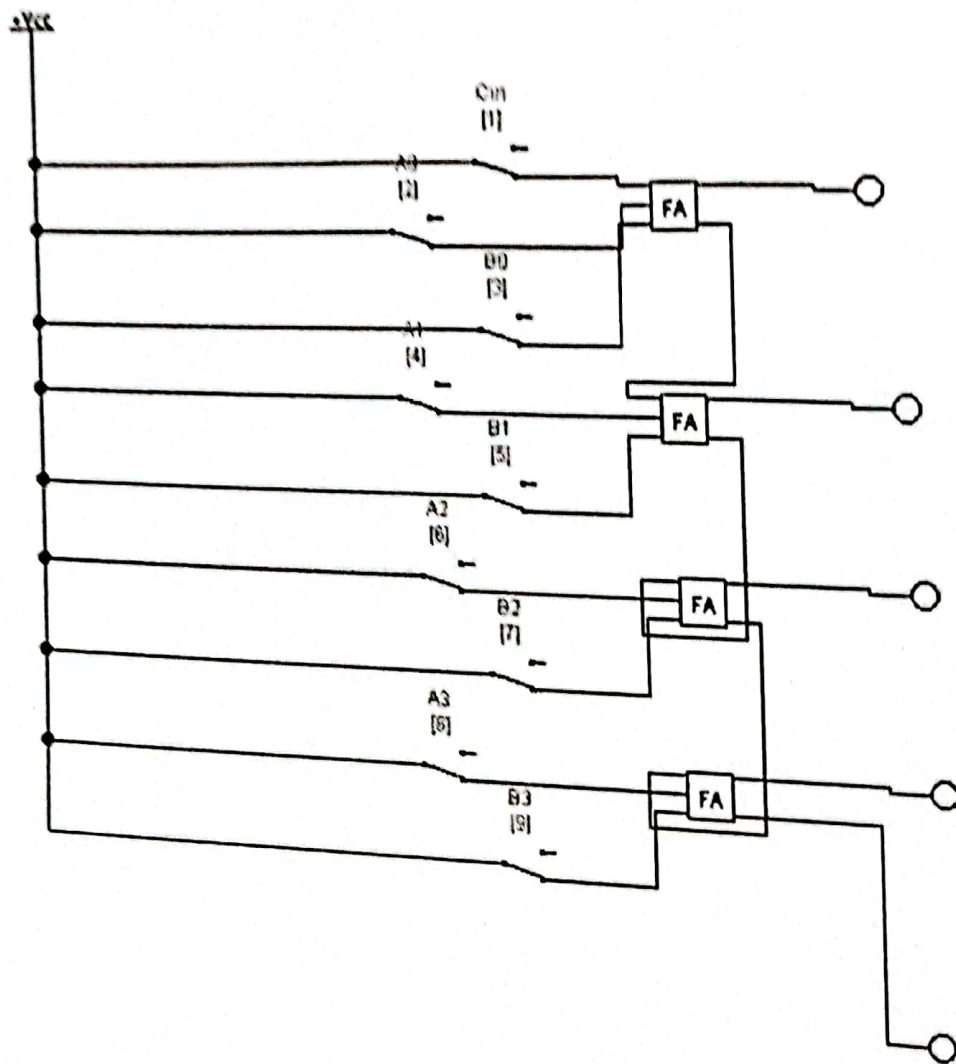
a. $A_3A_2A_1A_0 = 1011$

$B_3B_2B_1B_0 = 1101$

C_4	S_3	S_2	S_1	S_0
1	1	0	0	0

Practical No.5:





C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7
0	0	0	0	1	1	1	1	1	1	1
0	0	1	1	0	1	1	1	1	1	1
0	1	0	1	1	0	1	1	1	1	1
0	1	1	1	1	1	0	1	1	1	1
1	0	0	1	1	1	1	0	1	1	1
1	0	1	1	1	1	1	1	0	1	1
1	1	0	1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	0

EXERCISES

1. What will be the binary values at the outputs, Y0 through Y7, of 74138 if:

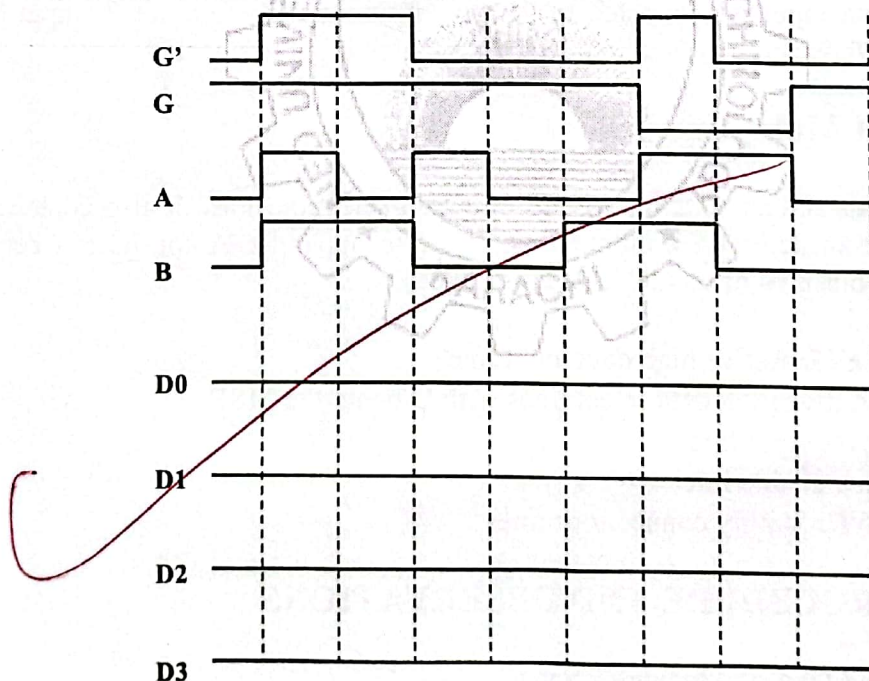
- All three enable pins are connected to ground?

In this case all outputs from Y0 to Y7 will be high since decoder will not be operating

- All three enable pins are connected to VCC?

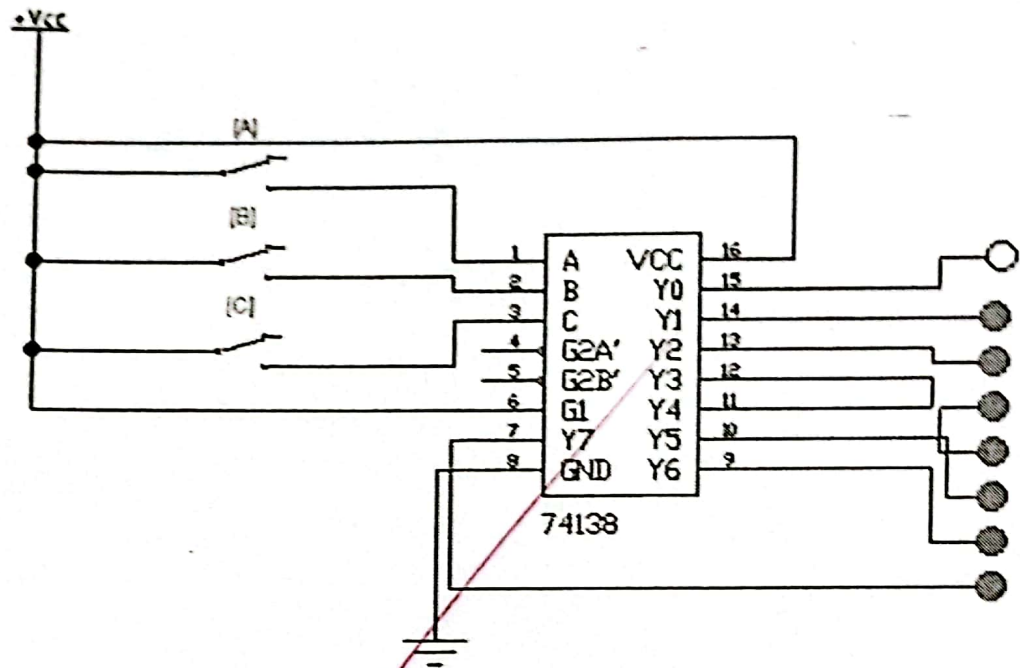
Again all outputs will be high as the decoder will be disabled

2. Consider a 2 x 4 Decoder with two enable inputs (one active high - G and one active low - G'). Draw the output wave-forms for D0, D1, D2 and D3, if the two select inputs are A and B (B being the MSB). All outputs are active low.

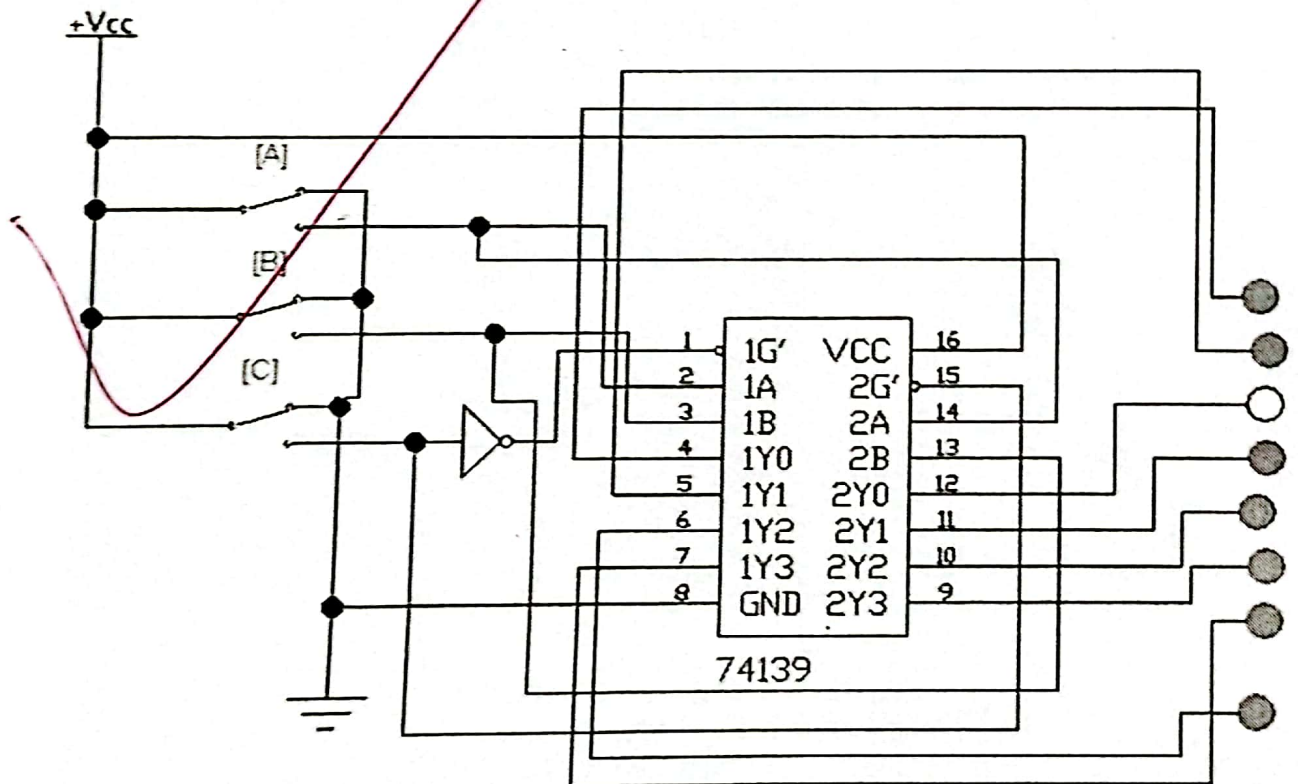


3. Cascade two 2x4 decoders to form a 3x8 decoder.

- Implement the circuit in hardware using two 74138 ICs and observe the output.
- Simulate the circuit on Electronics Workbench or any other simulation software using 74139 – dual 2-line-to-4-line decoder/demultiplexer IC. Attach hardcopy of the simulated circuit/output here.



Practical No.6(Q3):

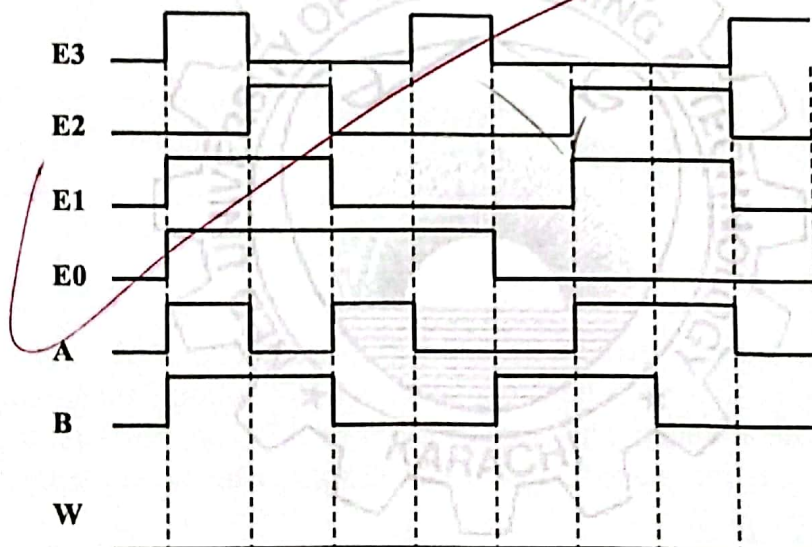


- 7 Record your observations for the input configurations given in the following table. Here only E0, E4 and E15 data inputs are considered.

G'	D	C	B	A	Value at data input	W	G'	D	C	B	A	Value at data input	W
0	0	0	0	0	1	0	0	1	0	0	1	0	1
0	0	0	0	1	0	1	0	1	0	1	0	0	1
0	0	0	1	0	0	1	0	1	0	1	1	0	1
0	0	0	1	1	0	1	0	1	1	0	0	0	1
0	0	1	0	0	1	0	0	1	1	0	1	0	1
0	0	1	0	1	0	1	0	1	1	1	0	0	1
0	0	1	1	0	0	1	0	1	1	1	1	1	0
0	0	1	1	1	0	1	1	0	1	0	1		
0	1	0	0	0	0	1	1	1	0	1	0		

EXERCISES

1. Consider 4x1 Multiplexer. Draw the output wave-form for the following data inputs (E₀, E₁, E₂, E₃,) and select lines A, B (B being the MSB). Output is W.

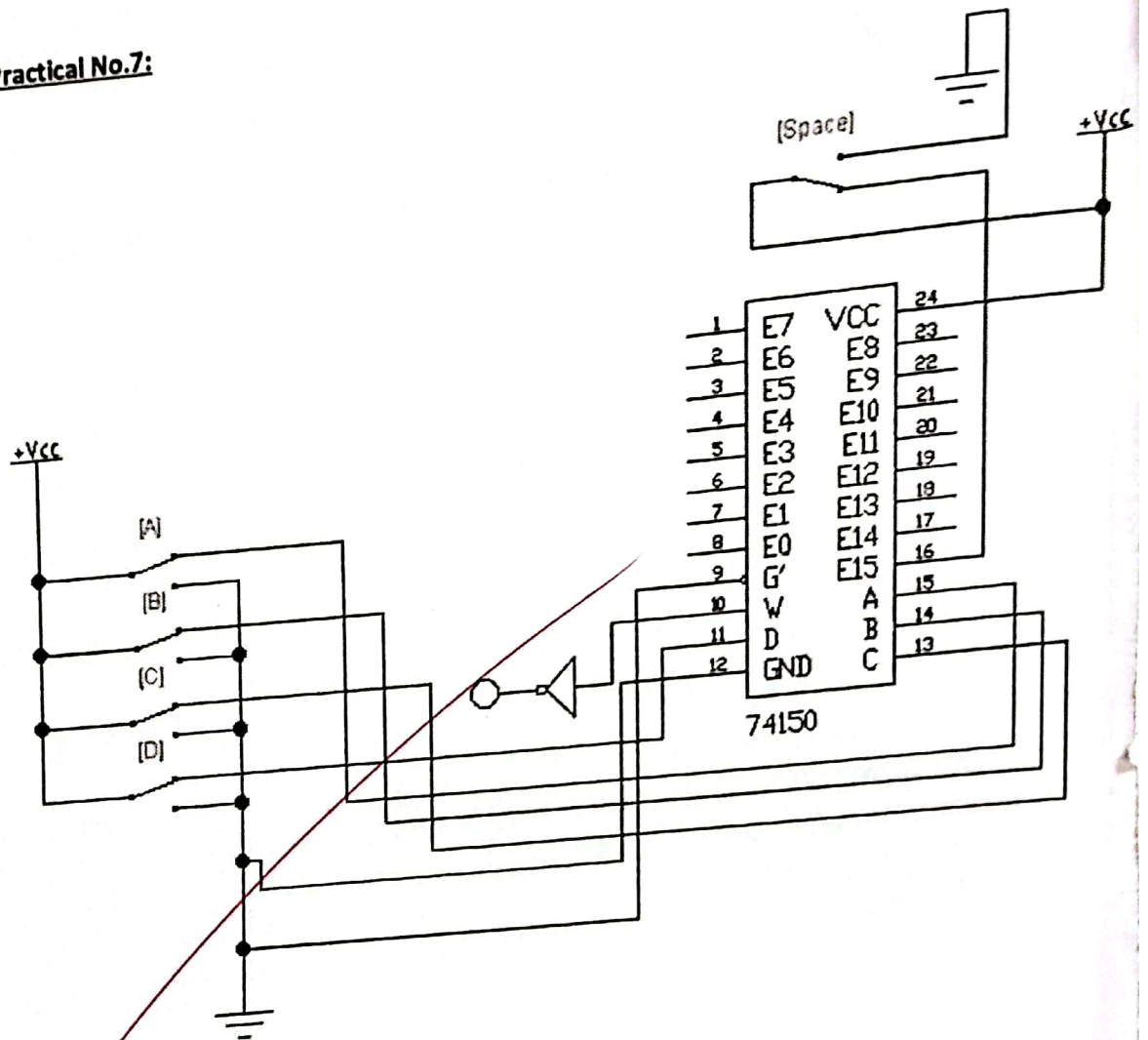


2. Implement the following function using 8x1 multiplexer.

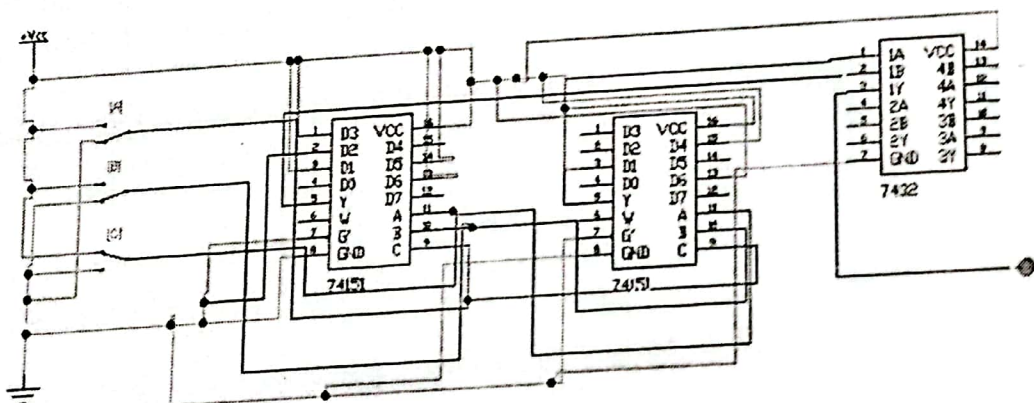
$$F(A, B, C, D) = \sum (1, 3, 5, 6, 9, 13, 15)$$

- Implement the circuit in hardware using 74150 IC and observe the output.
- Simulate the circuit on Electronics Workbench or any other simulation software using 74151 – 8-line-to-1-line multiplexer IC. Attach hardcopy of the simulated circuit/output here.

Practical No.7:



Practical No.7(Q2):



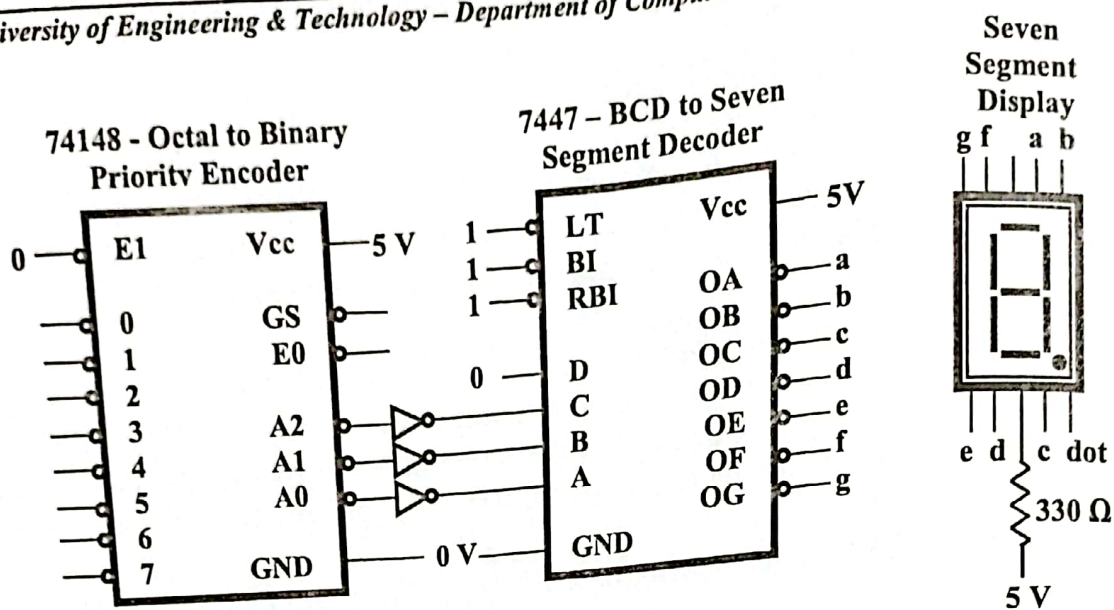


Figure 2: Circuit for displaying encoders output on 7-segment display

EXERCISES

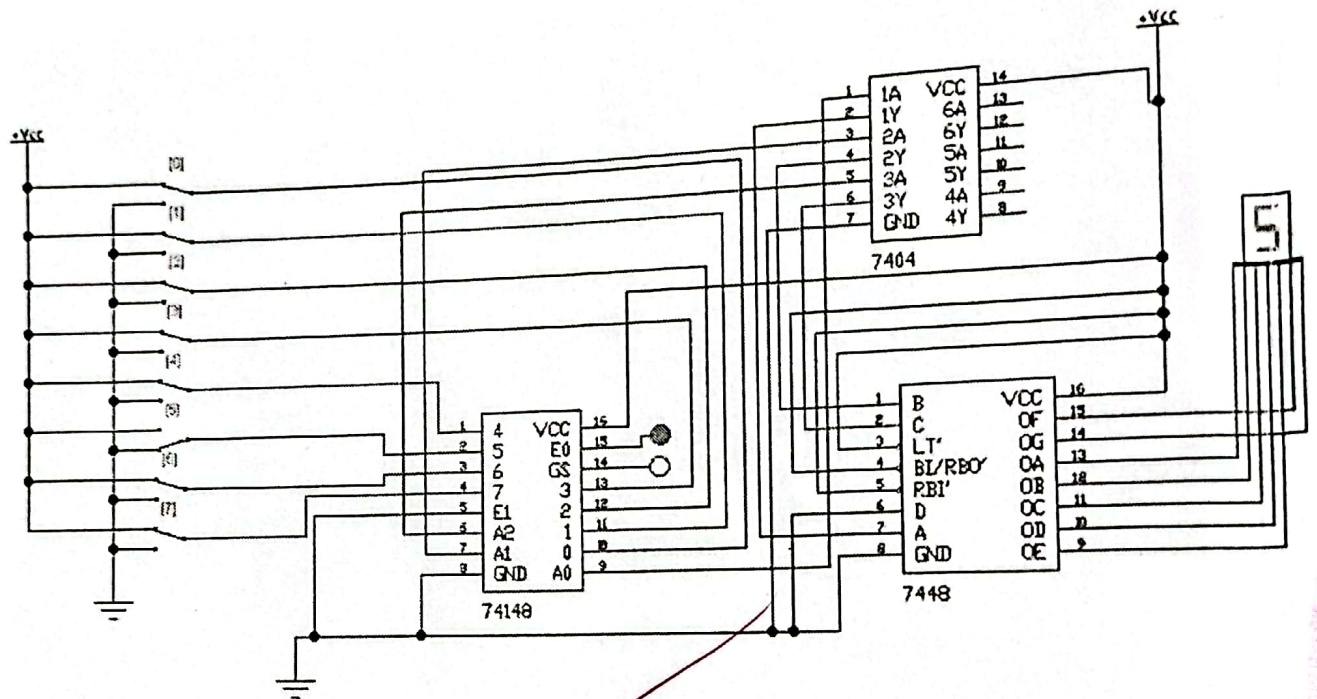
1. Perform the Lamp Test for the designed circuit and write your observations.

when the LT (Lamp Test) pin is given a low input, all segments of the 7 segment display will turn ON. This happens because the LT pin forces all outputs of the 7447 IC to go Low, which causes the display to light up all segments. It is done to check the working of display.

2. How can you use 7447 IC to drive a common-cathode display?

The 7447 IC is designed for common anode displays. To use it with a common cathode display, you would need to invert all of its outputs, because the outputs of the 7447 are active Low, while common cathode displays require active high inputs to light up the segments.

Practical No.8:



Lab Session 09

OBJECTIVE

Testing different modes of JK flip flop and implementing a modulo-4 asynchronous up-counter using JK flip flops

COMPONENTS AND APPARATUS REQUIRED

1. Digital logic trainer board or the following components:
Bread board, 5 V - Power Supply, Multimeter, LEDs with Resistors, Switches
2. Logic probe
3. Connecting wires
4. Following ICs and their datasheets or pin configurations:
5. 7473 / 7476 JK Flip-Flop

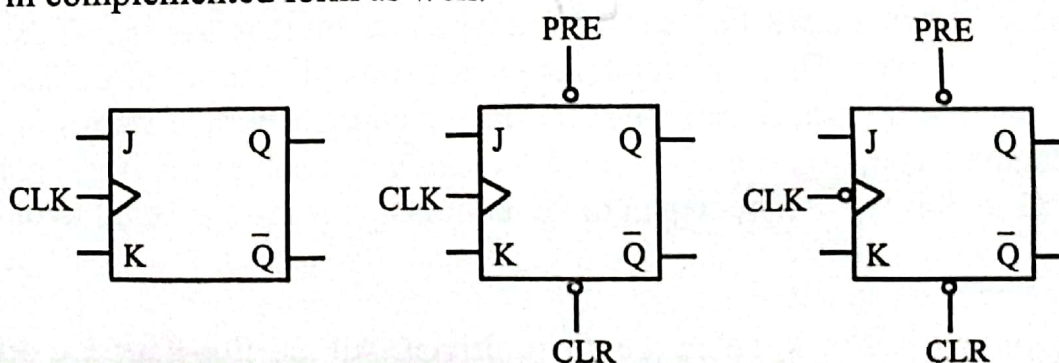
THEORY

Flip-Flop

A flip-flop circuit can maintain a binary state indefinitely (as long as the power is delivered to the circuit) until directed by an input signal to switch states. The major differences among various types of flip-flops are in the number of inputs they possess and in the manner in which the inputs affect the binary state.

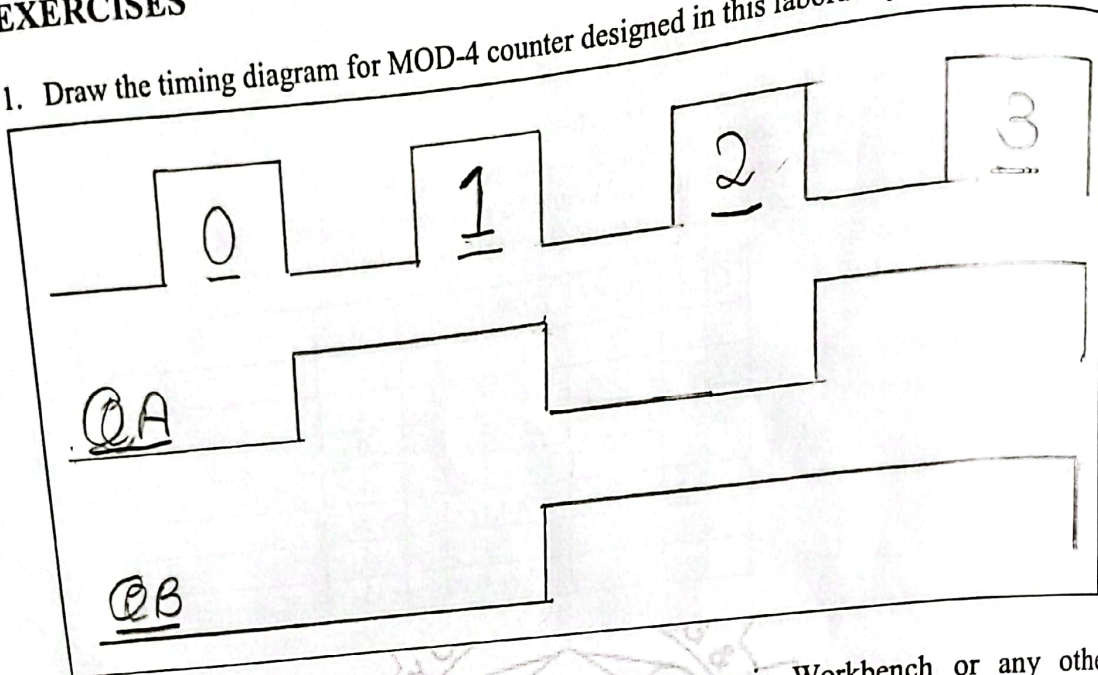
JK Flip-Flop

JK flip flop is an edge triggered device. A typical JK flip flop has three inputs: J, K and a clock input. The flip-flop can be either positive or negative edge triggered. The output Q is available in complemented form as well.



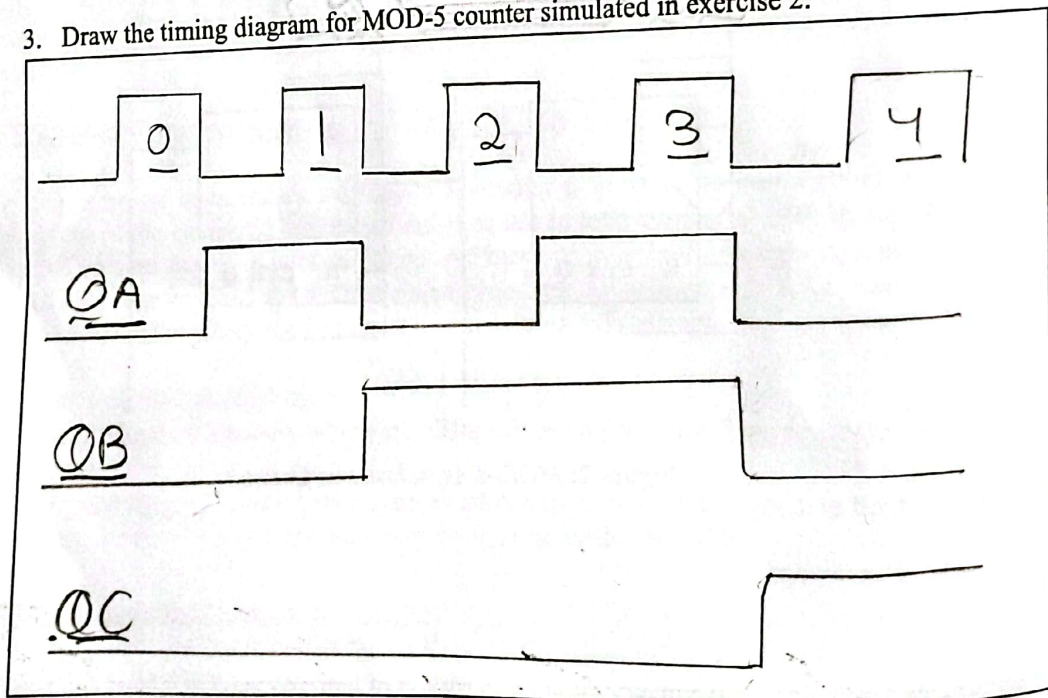
EXERCISES

1. Draw the timing diagram for MOD-4 counter designed in this laboratory session.

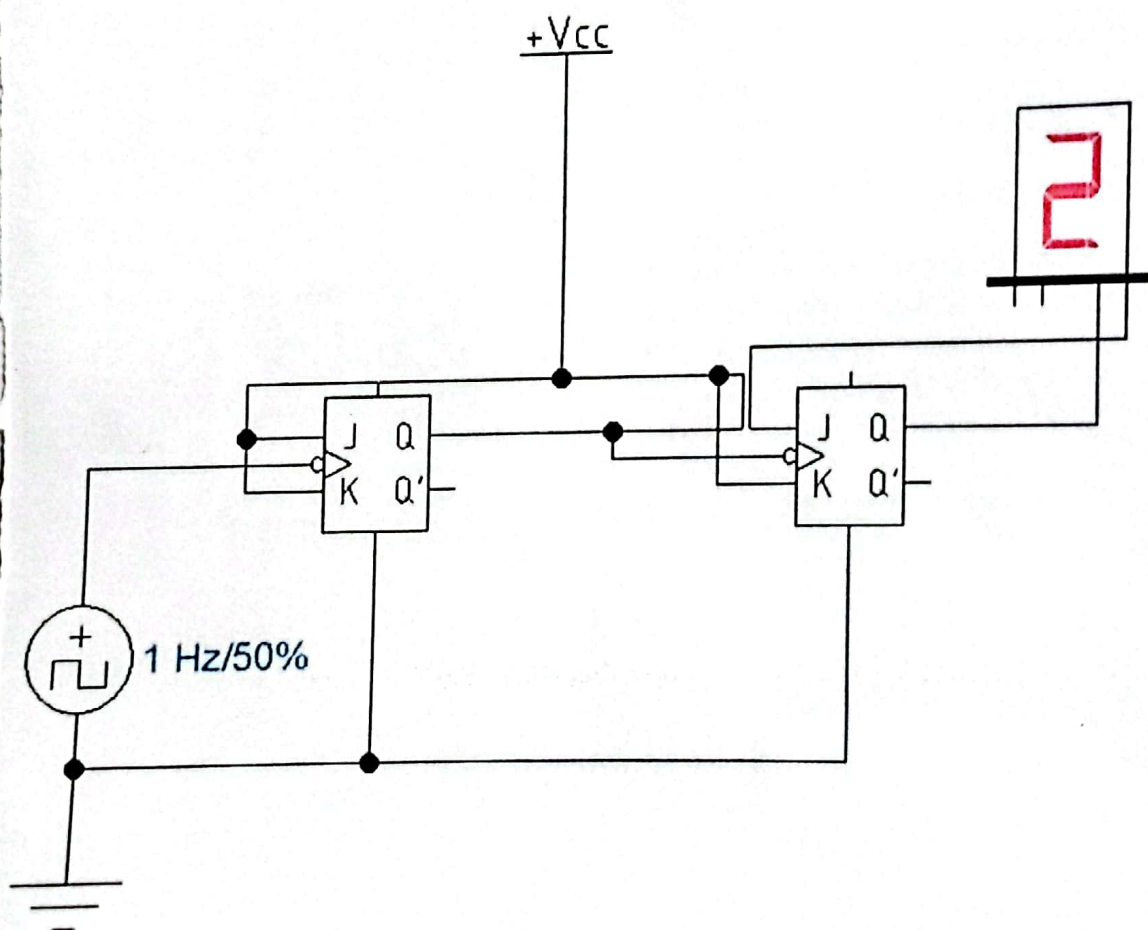


2. Simulate MOD-5 asynchronous counter using Electronic Workbench or any other simulation tool. The number of flip-flops required to construct a MOD-5 counter is 3. This counter will count from 0 to 4, a total of 5 distinct states. Since a 3-stage counter can count up to 8 states at maximum, a NAND gate is used to reset it after 5 clock pulses. Attach hardcopy of the simulated circuit/output here.

3. Draw the timing diagram for MOD-5 counter simulated in exercise 2.



LAB 09



Lab Session 10

OBJECTIVE

Experimenting with decade counter IC.

COMPONENTS AND APPARATUS REQUIRED

1. Digital logic trainer board or the following components:
Bread board, 5 V - Power Supply, Multimeter, LEDs with Resistors, Switches
2. Logic probe
3. Connecting wires
4. Following ICs and their datasheets or pin configurations:
5. 7490 decade counter

THEORY

7490 Decade Counter

7490 decade counter can generate the following two sequences:

1. BCD sequence: In this case output QA is connected to input B, external clock is applied to input A
2. Bi-quinary sequence: In this case output QD is connected to input A, external clock is applied to input B

Refer to appendix A for IC pin configuration. Function of various pins of this IC is described below:

- **Input A, Input B:** Negative edge triggered clock inputs.
- **QA, QB, QC and QD:** Active high outputs.
- **R91, R92, R01, R02:** Active high reset inputs. These resets are activated as follows:
- **VCC and GND:** Supply connections lines.

Reset Inputs				Output			
R01	R02	R91	R92	QD	QC	QB	QA
H	H	L	X	L	L	L	L
H	H	X	L	L	L	L	L
X	X	H	H	H	L	L	H
X	L	X	L	COUNT			
L	X	L	X	COUNT			
L	X	X	L	COUNT			
X	L	L	X	COUNT			

- Connect input A and input B according to the connections indicated in the previous section to generate BCD count and then bi-quinary count.
4. Observe the output in each case and record your observations given in the following table.

BCD Count				
Clock Pulse	QD	QC	QB	QA
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	0	0	0	0

Bi-quinary Count				
Clock Pulse	QD	QC	QB	QA
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	0
4	1	0	0	0
5	0	0	0	1
6	0	0	1	1
7	0	1	0	1
8	1	1	1	1
9	1	0	0	1
10	0	0	0	0

EXERCISES

1. Enable the RESET inputs R01 and R02 (keeping R91 and R92 disabled) and write your observations below:

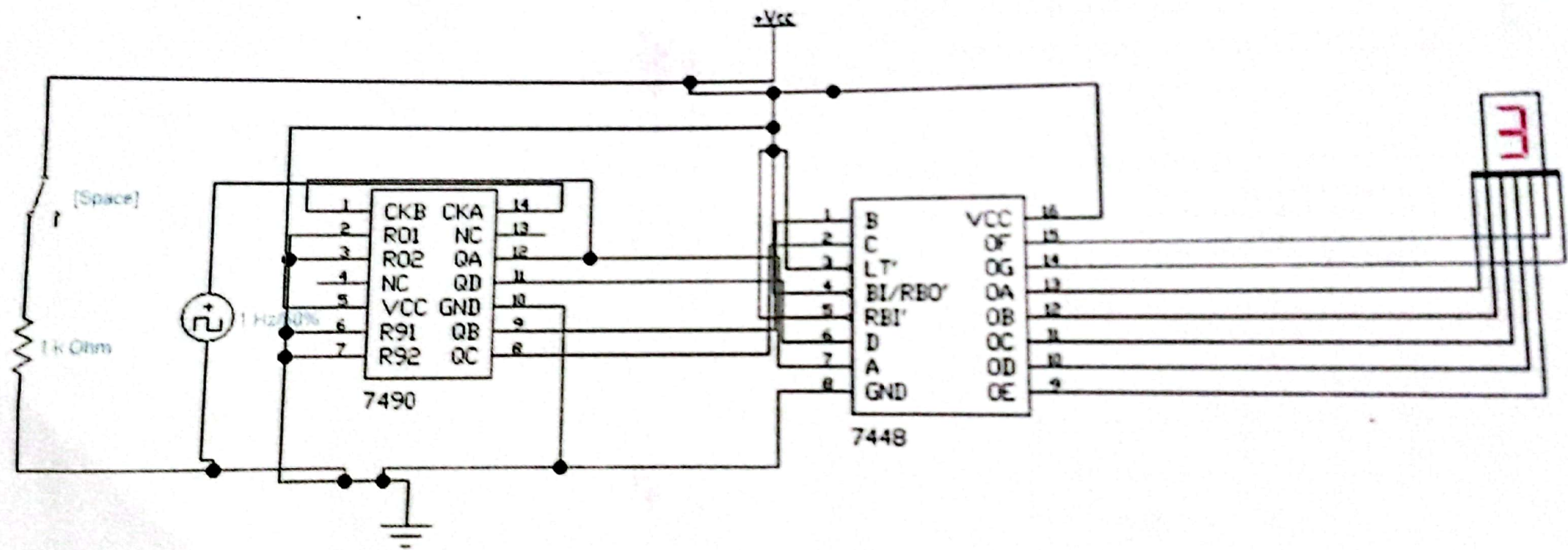
By enabling the RESET inputs, R01 and R02, the output of both BCD count and Bi-quinary Count at QA, QB, QC and QD is 0000.

2. Enable the RESET inputs R91 and R92 (keeping R01 and R02 disabled) and write your observations below:

By enabling the RESET input R91 and R92, the output of both BCD count and Bi-quinary count at QA, QB, QC and QD is 1001.

3. Enable all the RESET inputs R01, R02, R91 and R92 and write your observations below:

By enabling all the RESET input R01, R02, R91 and R92, the output of both BCD count and Bi Quinary count At QA, QB, QC and QD is 1001.



Lab Session 11

OBJECTIVE

Designing and simulating the seconds section of a digital clock

THEORY

Digital Clock

A digital clock is a time keeping circuit that displays seconds, minutes and hours. For seconds' section of a digital clock, a MOD-60 counter needs to be designed. A simple way to do this is to design MOD-10 and MOD-6 counters separately and then cascade them.

Generating Clock (Square Wave) Signals for Sequential Circuits in Electronic Workbench

Most sequential circuits require trigger/pulse for the functioning of their memory elements. This trigger is merely a 0 to 1 or 1 to 0 signal transition at the input where it needs to be applied. This can be achieved by simply connecting a switch which can toggle between 0 and 1.

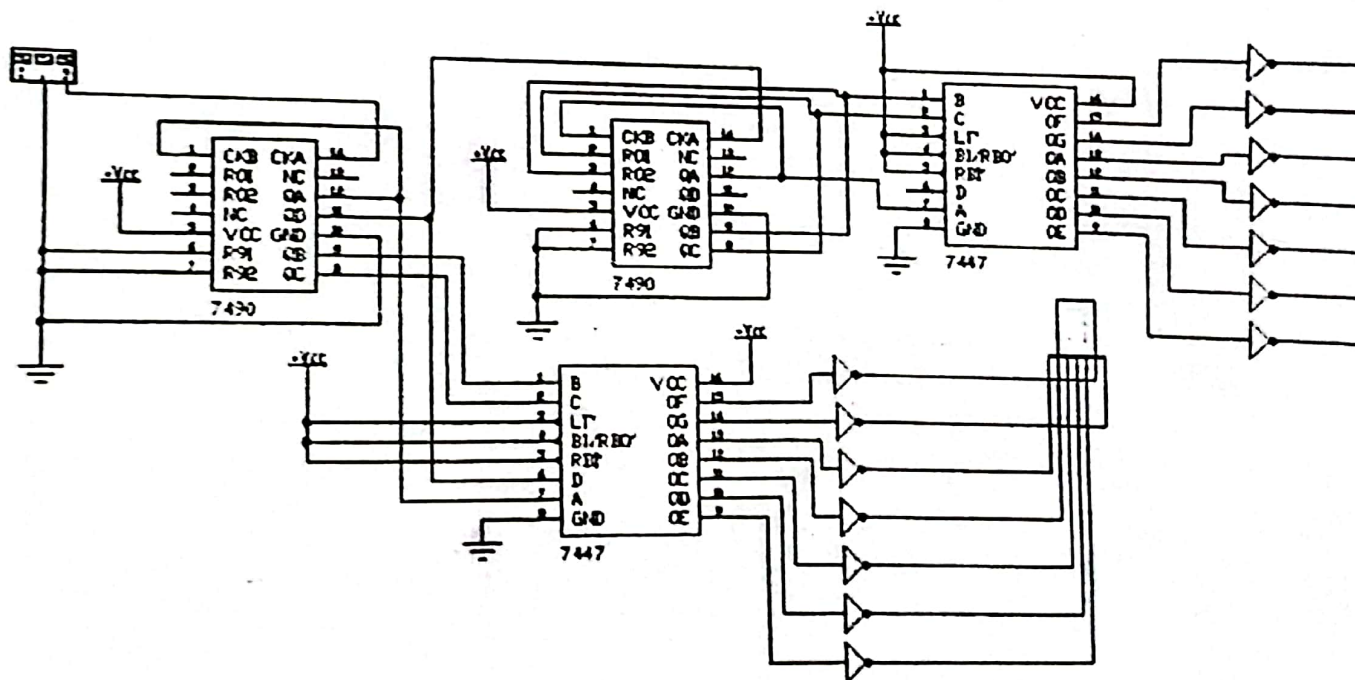
For continuous pulse generation a proper clock source is needed. In EWB, this can be achieved via function generator (in *Instruments Parts Bin*) or more easily as follows:

1. From the *Sources Parts Bin*, drag and drop *Clock* on the design area. Double click the *Clock* to reveal the properties dialog box, where frequency, duty cycle and voltage of the signal can be chosen.
2. Connect the negative end of the component to the ground.
3. Positive end can be connected to the point where the clock signal needs to be applied. You can connect an indicator here to view the generated signal.

PROCEDURE

In this lab session we will use two 7490 ICs to design MOD-10 and MOD-6 counters.

1. For MOD-10 counter, connect a 7490 IC in BCD count mode refer to previous lab session. Disable all RESET inputs and apply clock at input A as illustrated is the previous section. Set the frequency of the clock to 1 Hz.
2. Connect another 7490 IC in BCD mode. To convert it to MOD-6 counter, connect R01 to QB and R02 to QC. This will activate the resets of 7490 when it reaches the count of 6. Disable the other two RESET inputs R01 and R02. Now connect QD of the MOD-10



Lab Session 12

OBJECTIVE

Experimenting with bidirectional universal shift register IC

COMPONENTS AND APPARATUS REQUIRED

1. Digital logic trainer board or the following components:
Bread board, 5 V - Power Supply, Multimeter, LEDs with Resistors, Switches
2. Logic probe
3. Connecting wires
4. Following ICs and their datasheets or pin configurations:
5. 74194 4-bit bidirectional universal shift register

THEORY

Shift Registers

A Register is a set of flip-flops used to store binary data. A register, which is capable of shifting its binary information either to the right or left, is called a shift register. The logical configuration of a register consists of a chain of flip-flops connected in cascade, with the output of one flip-flop connected to the input of next flip-flop. All flip-flops receive a common clock pulse, which causes the shift from one stage to the next.

Universal Bidirectional Shift Registers

A bi-directional shift register is one whose bits can be shifted from left to right or from right to left. A universal shift register is a bi-directional register whose input can be in either serial or parallel form and whose output can be in either serial or parallel form.

74194 4-Bit Bidirectional Universal Shift Register

The 74194 register provides parallel as well as serial loading in both directions. A, B, C, and D are inputs for parallel loading, whereas SR and SL are inputs for serial loading with right or left shifting respectively. S1 and S0 are used to select the loading mode. Refer to appendix A for IC pin configuration. Function of various pins of this IC is described below:

- **A, B, C and D:** Active high inputs for parallel loading.
- **QA, QB, QC and QD:** Active high outputs.
- **S0 and S1:** Active high mode control inputs. The following table shows combinations of S1 and S0 to enable various modes.

- **SR:** Active high serial input for right shifting.
- **SL:** Active high serial input for left shifting.
- **CLR:** Active low clear input.
- **CLK:** Positive edge triggered clock input.
- **VCC and GND:** Supply connections lines.

TESTING PROCEDURE AND OBSERVATIONS

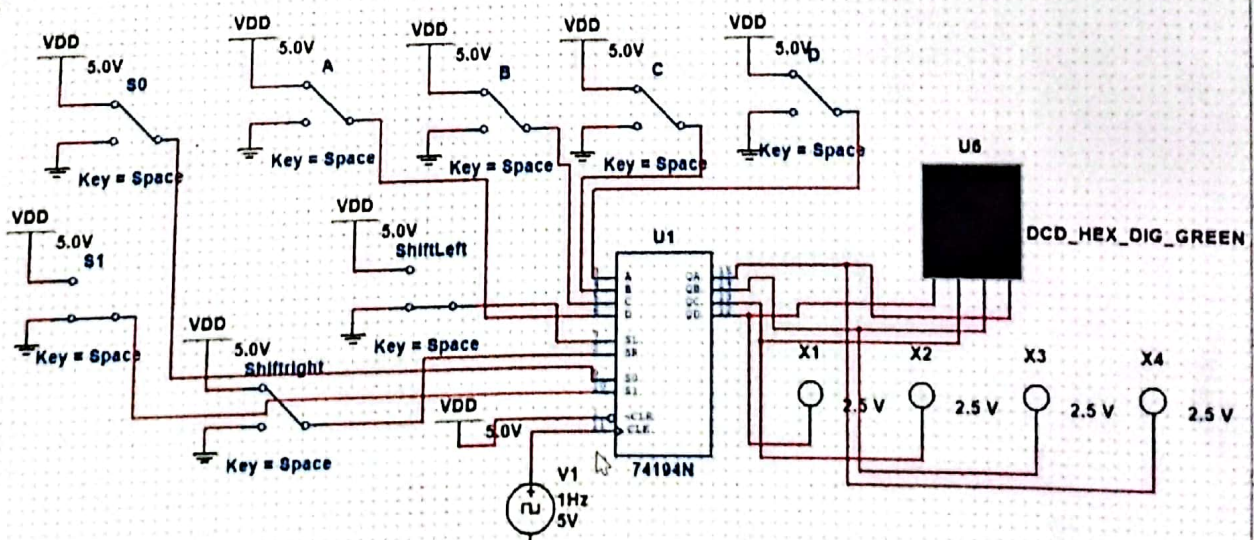
- 1 Make Vcc and Gnd supply connections.
- 2 Connect the CLK input to some clock source or a switch.
- 3 Disable CLR input.
- 4 Apply different combinations of 1s and 0s at inputs J, K, Preset and Clear.
- 5 Observe the output and record your observations given in the following table.

SI	SO	SL	SR	A	B	C	D	QA	QB	QC	QD
0	0	0	1	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	1	0	0	0	1
1	0	0	0	0	0	1	0	0	1	0	0
1	1	0	0	0	0	1	1	0	0	1	1
0	0	1	1	0	1	0	0	0	0	1	1
0	1	1	1	0	1	0	1	0	0	0	1
1	0	1	0	0	1	1	0	1	0	1	1
1	1	1	0	0	1	1	1	0	1	1	1
1	1	0	0	1	1	1	1	1	1	1	1

EXERCISES

1. Show connections of 74194 to convert it into a ring counter with right shifting.

1	CLR'	Vcc	16
2	SR	QA	15
3	A	QB	14



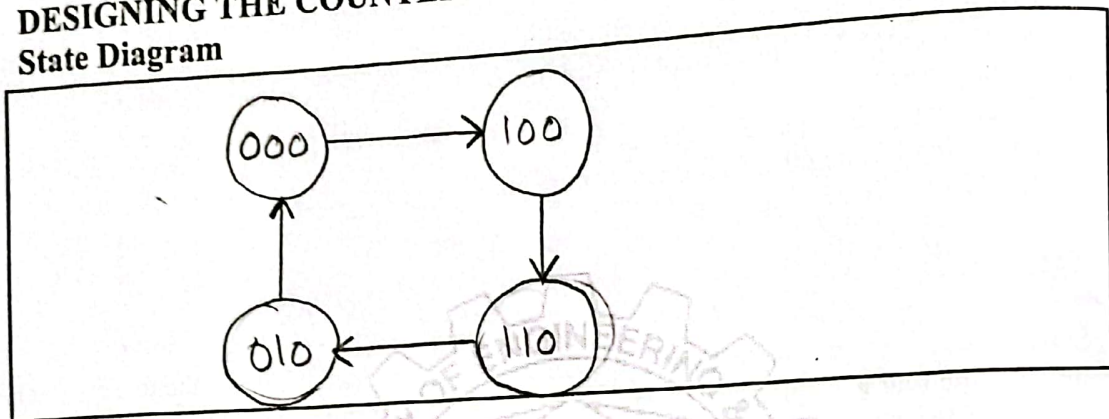
T flip-flop (toggle flip flop) has a single input T. If the T input is high, the T flip-flop changes state (toggles) whenever the clock input strobes. If the T input is low, the flip-flop holds the previous value. This behavior is described by the characteristic table and transition table given in figure 1.

Characteristic equation of T flip-flop is:

$$Q_{(t+1)} = T \oplus Q_t$$

DESIGNING THE COUNTER

State Diagram



State Table

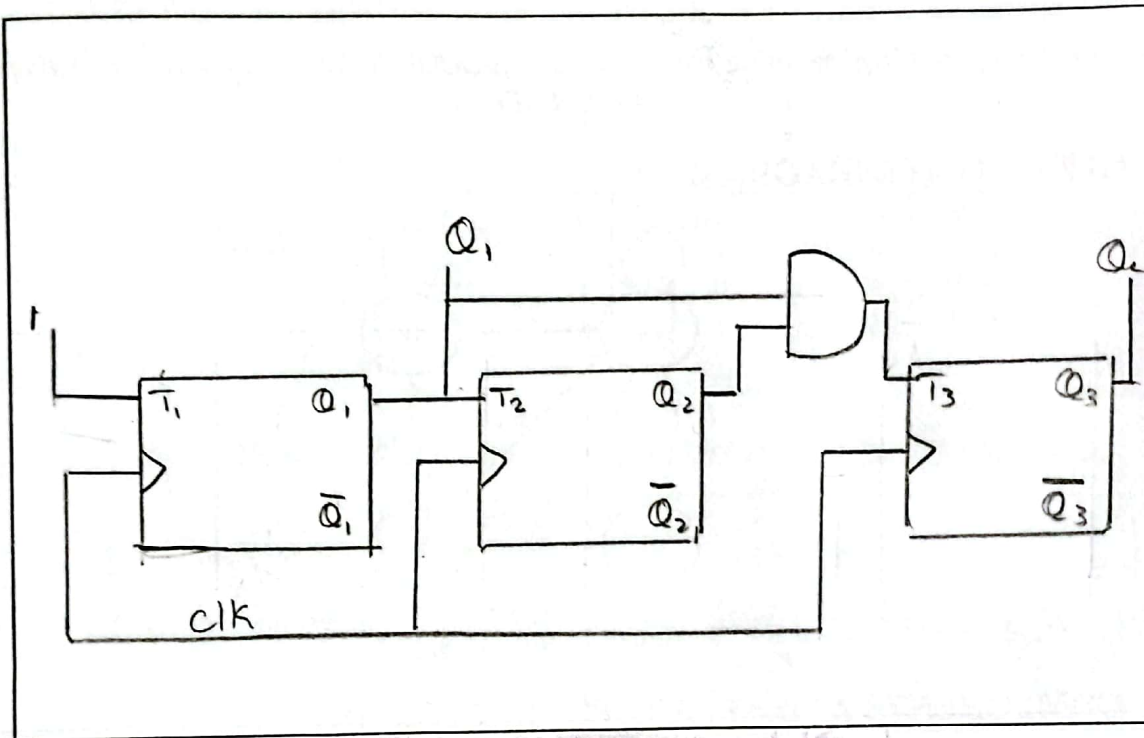
The design requires three flip flops: FF-0, FF-1 and FF-2.

Present State			Next State			Flip flop inputs		
Q_{2t}	Q_{1t}	Q_{0t}	$Q_{2(t+1)}$	$Q_{1(t+1)}$	$Q_{0(t+1)}$	FF-2, T_2	FF-1, T_1	FF-0, T_0
0	0	0	0	1	0	0	1	0
0	0	1						
0	1	0	1	0	0	1	1	0
0	1	1						
1	0	0	1	1	0	0	1	0
1	0	1						
1	1	0	0	0	0	1	1	0
1	1	1						

Input Equations

$T_2 = 0$, $T_1 = 1$, $T_0 = 0$

Circuit Diagram



1. Implement the circuit using digital ICs on a bread board (refer to appendix A for IC pin configurations). T flip flop can be obtained by connecting together the two inputs J and K, of JK flip flop.
2. Make Vcc and Gnd supply connections.
3. Connect the CLK input to some clock source or a switch.
4. Observe the output even value sequence of the circuit.

D Flip-Flop

D flip-flop is also called transparent flip-flop as it simply transfers the input data to the output. This behavior is described by the characteristic table and transition table given in figure 2.

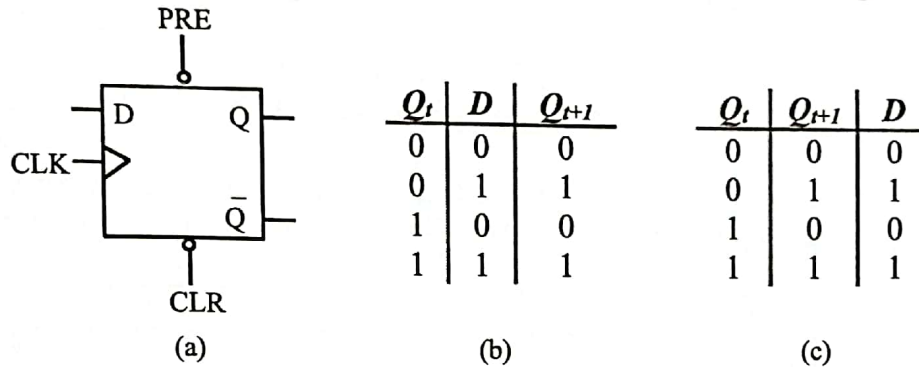


Figure 2: D Flip Flop. (a) Graphical symbol. (b) Characteristic table. (c) Excitation table

Characteristic equation of D flip-flop is: $Q_{(t+1)} = D$

DESIGNING SYNCHRONOUS SEQUENTIAL CIRCUIT FOR THE GIVEN STATE DIAGRAM

State Table

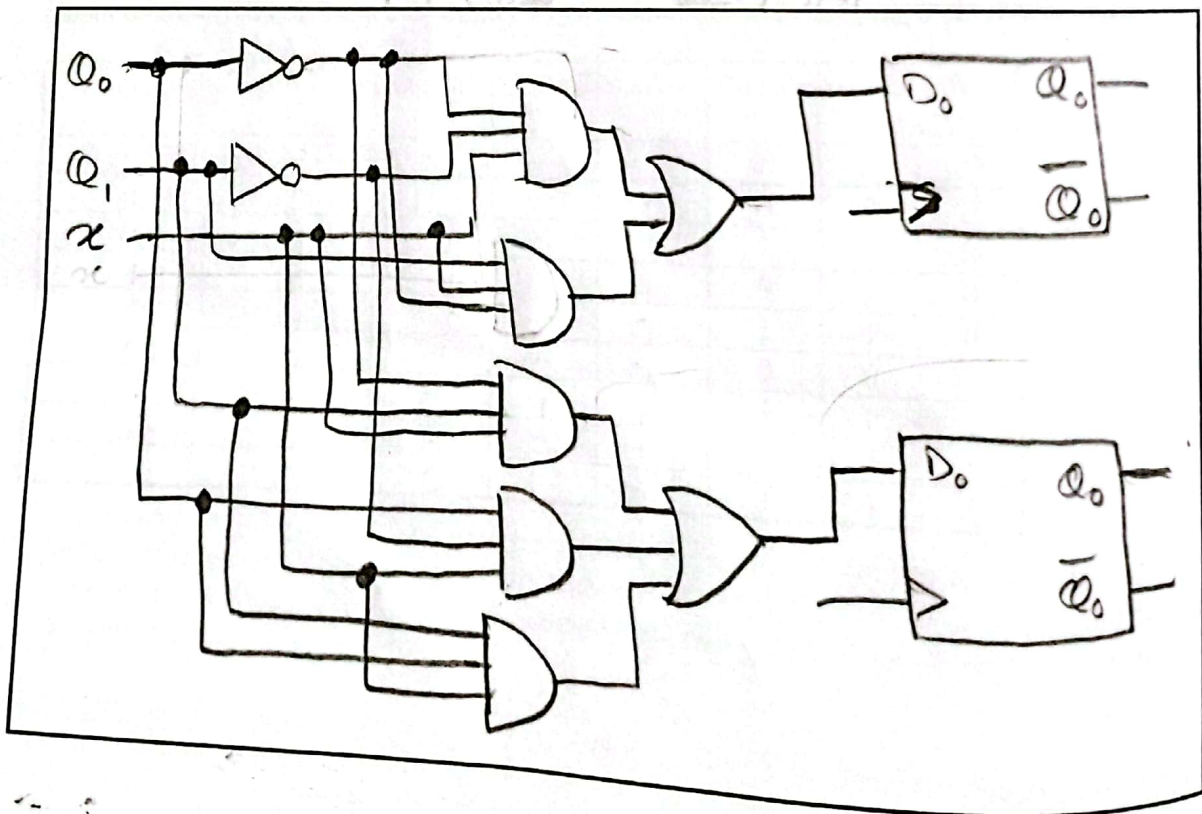
The given state diagram has four states, so two D flip-flops (FF-0, FF-1) will be required.

Present State		Input	Next State		Output	Flip flop inputs	
Q_{1t}	Q_{0t}	x	$Q_{1(t+1)}$	$Q_{0(t+1)}$	z	FF-1, D_1	FF-0, D_0
0	0	0	0	1	1	0	1
0	0	1	1	0	0	1	0
0	1	0	1	1	1	1	1
0	1	1	0	0	0	0	0
1	0	0	0	0	1	0	0
1	0	1	1	1	0	1	1
1	1	0	1	0	1	1	0
1	1	1	0	1	0	0	1

Input and Output Equations

$$D_1 = Q_1 \oplus Q_0 \oplus x, D_0 = \overline{Q_1} \cdot \overline{Q_0} \cdot x + Q_1 \cdot Q_0 \cdot x, Z = \overline{Q_1} \cdot \overline{Q_0} + \overline{Q_1} \cdot Q_0 + Q_1 \cdot \overline{Q_0}$$

Circuit Diagram



IMPLEMENTATION PROCEDURE AND OBSERVATIONS

1. Implement the circuit using digital ICs on a bread board (refer to appendix A for IC pin configurations).
2. Make Vcc and Gnd supply connections.
3. Connect the CLK and the CLR inputs to input switches (preferably push buttons).
4. Apply 0 to the input x.
5. Enable CLR input momentarily to clear all the flip flops.
6. Apply CLK through the switch and record your observations in the following table.
7. Apply 1 to the input x.
8. Enable CLR input momentarily to clear all the flip flops.
9. Apply CLK through the switch and record your observations in the following table.

$x = 0$				$x = 1$			
Clock Pulse #	Q_1	Q_0	Output z	Clock Pulse #	Q_1	Q_0	Output z
Initial value	0	0	1	Initial value	0	0	1
1	0	1	1	1	1	0	0
2	1	1	0	2	1	1	0
3	1	0	0	3	0	1	1
4	0	0	1	4	0	0	1
5	0	1	1	5	1	0	0

Alternatively, connect the CLK input to a continuous clock source. Now output will change continuously. These outputs can be viewed using an oscilloscope.